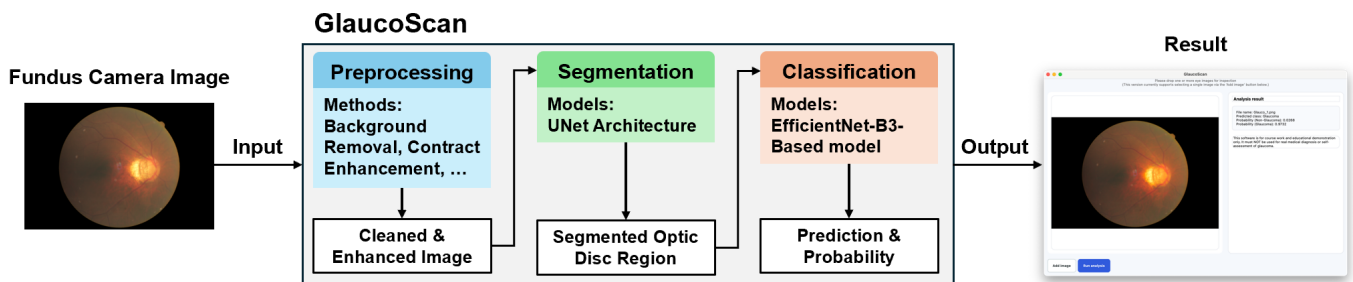


👁️ GlaucoScan

Overview

This project is a deep learning-based application designed to assist in the preliminary analysis of **Fundus Camera Images** to detect the signs of **Glaucoma**.

GlaucoScan's Three-stage Pipeline



Features

- **Targeted Diagnostic Focus:** Combines robust **preprocessing** and **UNet segmentation** to ensure the classifier focuses exclusively on the critical optic disc region, maximizing diagnostic accuracy.
- **High-Performance Backbone:** Employs a fine-tuned **EfficientNet-B3 model** to achieve reliable and efficient feature extraction for binary glaucoma classification.
- **Fundus Image–Optimized Processing:** The pipeline is custom-tailored to neutralize the visual noise (dark backgrounds, contrast variation) inherent in **diverse clinical fundus images**.
- **End-to-End Automated Workflow:** Provides a fully integrated pipeline from image input to prediction output, minimizing manual intervention.
- **User-Oriented Graphical Interface:** Offers an intuitive GUI that allows non-technical users to perform glaucoma analysis with minimal effort.

🔧 Installation and Setup

This software offers two primary methods for usage: running the analysis via **Python scripts** or utilizing the comprehensive **standalone desktop application** built with PyQt5.

📁 Source Code Installation (Python Environment)

1 Prerequisites

- Python 3.8+
- NVIDIA GPU (Recommended for faster inference, otherwise runs on CPU)

2 Environment Setup

2.1 Clone the repository

```
git clone https://github.com/Brokengm/BIA_group_work.git
cd final_version
```

2.2 Create and activate the environment

```
# Using conda
conda create -n glaucoma python=3.9
conda activate glaucoma
```

💡 Pro Tip: We highly recommend using a **Conda environment** for dependency management. Conda simplifies package isolation and installation, especially for scientific and deep learning packages like PyTorch.

🔗 Further Reading: [Conda Documentation](#) | [Managing Conda Environments](#)

2.3 Install dependencies

```
pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu118 # Use appropriate CUDA version
pip install -r requirements.txt
# The required packages include: PyQt5, numpy, Pillow, timm, scikit-image
```

2.4 Prepare files

For the application to run, you must download the these files and place them directly into the same directory:

- [384_unet.pth](#) UNet Segmentation Model
- [efficientnet.pth](#) EfficientNet Classification Model
- [app_gui.py](#) Main application, GUI (PyQt5) logic, and prediction pipeline entry point.
- [model.py](#) Contains UNet and CombinedModel (EfficientNet) class definitions and model loading logic.
- [preprocess.py](#) Contains all image processing functions (cropping, CLAHE, segmentation, etc.).
- [welcome.jpeg](#) The picture of main page.

3 Execute the script

Ensuring you have the 6 files prepared, run the main application script under the same directory.

```
python app_gui.py
```



Download [GlaucoScan.zip](#), and decompress it to run the software immediately.

Analysis Steps

1. **Welcome Screen:** Click "**Let's get started**".
2. **Load Image:** Click the "**Add image**" button to select a fundus image (.jpg, .png, etc.). The image will appear in the left panel.
3. **Analyze:** Click "**Run analysis**". The system will perform:
 - Preprocessing (border removal, contrast enhancement).
 - Optic Disc Segmentation (using UNet).
 - Optic Disc Cropping (centering on the segmented region).
 - Glaucoma Classification (using EfficientNet-B3).
4. **View Results:** The classification result, including the predicted class (Glaucoma or Non-Glaucoma) and the precise probability scores, will be displayed in the right panel.

A visual walkthrough of these analysis steps is available in the [Introduction Video](#).

Disclaimer

This software is intended for course work and educational demonstration only. It must NOT be used for real medical diagnosis or self-assessment of glaucoma. Always consult a qualified medical professional for health concerns.